

# Feature Expansion Integration Plan

Prepared: June 2026 | Scope: 15 families × 15 merge stages + sub-object chains + power-up consumables

Audience: Implementation sessions — assumes full codebase familiarity

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## A. Current State Assessment

### How animal families/chains are currently defined

**AnimalSpecies.swift** is the authoring source. The *AnimalSpecies* enum has 15 cases (.dog, .cat, .rabbit, .bird, .hamster, .turtle, .fox, .owl, .fish, .lizard, .ferret, .parrot, .pony, .hedgehog, .guineaPig), each declaring:

- tierNames: [String] — **exactly 9 entries** per family, one per RescueStage
- spawnerName, spawnerSFSymbol, sfSymbol, tintColor

**ItemChain.swift** builds the live chain objects. *ContentRegistry.makeAnimalChain()* zips *RescueStage.allCases* (9 cases) with *AnimalSpecies.tierNames* (9 entries) to produce a 9-tier MergeChain. The chain ID is "animal.(species.rawValue)" — a stable string stored in saves.

**Implication:** The current chain length (9) is structurally enforced by *RescueStage.allCases.count*. Expanding to 15 requires decoupling tier count from *RescueStage*.

### How the merge board works

- **Grid:** 9 rows × 7 cols = 63 cells. Rows 7 and 8 start locked (unlock at levels 3 and 8).
- **BoardCell** holds either a BoardItem? (merge tile) or a ProducerTile? (spawner), never both.
- **BoardItem(chainID, tier)** is the only thing persisted. Display data is looked up from ContentRegistry at runtime.
- **Merge logic** (attemptMergeOrMove): two items with the same chainID and tier merge into tier+1. Top tier check: next == chain.maxTier. This is fully data-driven — extending to 15 tiers requires only a new chain definition.
- **Spawner model:** ProducerLevel.familySpawner (rawValue 20) costs kibble, has no charges, always drops the family's animal chain. No sub-object drop logic yet.
- **Top-tier celebration** fires in triggerTopTierCelebration. Awards coins, a toolbox, increments ambassadors, triggers MilestoneManager.

### How inventory is currently structured

- Animal inventory: 18 slots (6 free + 2 × 6 purchasable rows). Items routed by chain.category == .animal.
- Tool/material inventory: 18 slots. Items routed by category == .tool or .material.
- Producer storage: one designated slot per ProducerLevel.rawValue + 4 overflow slots.
- No tab-based separation exists in code — the two tabs are visual only (share InventoryStore). GDD v2.0 mentions a 3-tab inventory but this is not yet implemented.

### How quests/orders currently work

- QuestGoal has four cases: .mergeAny, .mergeInChain(chainID, count), .reachTier(chainCategory, tier, count), .spawnBase.
- .reachTier(.animal, tier: N, count: C) uses RescueStage(rawValue: N+1) for labels — only resolves cleanly for tiers 0–8.
- AdoptionBoard.generateOrder uses RescueStage cases directly; maxAchievableOrderTier clamps to tier 8.

- AdoptionOrder.orderDescription and stageBadgeSymbol call QuestGoal.animalTierSymbol(tier), which returns nil/fallback for tiers 9–14.

## How map expansion currently works

SanctuaryMap.swift defines 6 areas in sequential unlock order:

| # | Area                 | Unlocks                                               |
|---|----------------------|-------------------------------------------------------|
| 1 | Antique Dog House    | Tutorial, no new spawner (Canines is day-one starter) |
| 2 | Scratching Post      | Unlocks .cat (Felines)                                |
| 3 | Garden Hutch         | Unlocks .rabbit (Lagomorphs)                          |
| 4 | Decorative Birdhouse | Unlocks .bird (Avians)                                |
| 5 | Wooden Burrow        | Unlocks .hamster (Rodents)                            |
| 6 | Heated Rock          | Unlocks .turtle (Reptiles)                            |

The remaining 9 families (Ursids, Aquatics, Amphibians, Marsupials, Primates, Equines, Pachyderms, Bovines, Cervids) have no map areas defined. **Note:** Aquatics (.fish) has a separate milestone unlock at 50 ambassador stars — this conflicts with adding a map area for Aquatics.

## Gaps and conflicts — current architecture vs. new requirements

| Gap / Conflict                                         | Location                                                       | Severity                            |
|--------------------------------------------------------|----------------------------------------------------------------|-------------------------------------|
| RescueStage (9 cases) enforces 9-tier chain length     | AnimalSpecies.swift, makeAnimalChain, AdoptionBoard, QuestGoal | Critical — must expand to 15        |
| tierNames only has 9 entries per family                | AnimalSpecies.swift                                            | Critical — must expand to 15        |
| maxAchievableOrderTier caps at tier 8                  | AnimalSpecies.swift                                            | High — breaks orders for tiers 9–14 |
| AdoptionBoard.generateOrder uses RescueStage weighting | AdoptionBoard.swift                                            | High — needs new weighting table    |
| QuestGoal.animalTierLabel/Symbol/Color fallbacks       | AnimalSpecies.swift, RescueStage                               | High — returns nil for tiers 9–14   |
| sellValue(forTier:) is a 9-entry array                 | MergeBoardViewModel.swift                                      | Medium — needs 15 entries           |
| No sub-object chain category                           | ItemChain.swift                                                | High — need new ChainCategory       |
| No drop logic in family spawner                        | MergeBoardViewModel.activateProducer                           | High — new spawn path needed        |
| No power-up consumable model                           | New file needed                                                | High                                |
| No pity timer state                                    | GameState                                                      | Medium — new fields needed          |
| 9 families have no map areas                           | SanctuaryMap.swift                                             | Medium — additive, no conflicts     |
| Aquatics has dual unlock paths                         | MilestoneManager, SanctuaryMap                                 | Medium — decision needed            |
| GameStore.currentVersion = 20 needs a backup           | GameStore.swift                                                | Low — additive migration only       |

## B. Data Model Plan

### B.1 AnimalSpecies.swift changes

**tierNames — expand from 9 to 15 entries per family.** Example for Canines:

```
["Pup", "Kit", "Houndling", "Terrier", "Spaniel", "Scout",  
"Retriever", "Shepherd", "Husky", "Alpha", "Guardian", "Sentinel",  
"Dire Wolf", "Mythic", "Primordial"]
```

All 15 families × 15 stages must be entered verbatim. Pure data edit — no logic changes required.

**maxAchievableOrderTier(forPlayerLevel:)** — replace RescueStage-based lookup with a direct tier-index table.

```
levels 1-3:   tier 2   (Houndling-equivalent)
levels 4-6:   tier 5   (Scout-equivalent)
levels 7-9:   tier 8   (Husky-equivalent)
levels 10-12: tier 10  (Guardian-equivalent)
levels 13-18: tier 12  (Sentinel-equivalent)
levels 19+:   tier 14  (top tier)
```

**RescueStage** — keep as-is but stop using it as authoritative tier count. Recommendation: add a standalone *animalTierAppearance(tier: Int)* function in *AnimalSpecies.swift* covering all 15 tiers, and replace all call sites of *QuestGoal.animalTierLabel/Symbol/Color* with it.

**sellValue(forTier:)** in *MergeBoardViewModel.swift* — extend to 15 entries. Current: [1,2,5,10,20,50,100,200,500]. Suggested extension: [..., 1000, 2000, 5000, 10000, 25000, 100000]. Exact values need sign-off (see Section E, Q3).

## B.2 ItemChain.swift changes

**ContentRegistry.makeAnimalChain()** — one line changes. Replace `zip(RescueStage.allCases, tierNames)` with `tierNames.enumerated()`. Score scale: `scoreValue = (index+1)*25`, `xpValue = (index+1)*xpPerMergeBase`. Badge moves from `stage == .ambassador` to `index == 14`.

**New ChainCategory cases:**

```
case subObject // per-family 4-stage merge chain → power-up on completion
case powerUp   // power-up consumable (drag onto spawner to apply)
```

**New stable ChainIDs:** pattern "subobject.(species.rawValue)" — e.g. "subobject.dog" for Canines (Biscuit → Bone → Chew Toy → Golden Ball).

**ContentRegistry.init()** — register sub-object chains:

```
for species in AnimalSpecies.allCases { register(Self.makeSubObjectChain(species)) }
```

## B.3 New file: SubObjectSystem.swift

Responsible for: (1) resolving spawner drops, (2) pity timer logic, (3) applying power-up effects. Key types:

- *SubObjectRarity* enum — .speed (60%), .mapSupplies (25%), .spawnerRefill (10%), .highTierDrop (5%)
- *PowerUpEffect* enum — 2x Speed/30s; Spawner Refill; High-Tier Drop Guarantee; Map Supplies
- *SpawnDropResult* enum — .animal(tier: Int) or .subObject(chainID: ChainID, tier: Int)
- *PityState* (Codable) — subObjectSpawnsSinceLastRare, subObjectSpawnsSinceLastEpic, pity thresholds

Key function signatures:

```
func resolveSpawnerDrop(species: AnimalSpecies,
                        pityState: inout PityState,
                        spawnTier: Int) -> SpawnDropResult

func applyPowerUp(effect: PowerUpEffect, to producer: inout ProducerTile)
```

## B.4 Summary: what changes vs. what is new

| File                      | Action | Notes                                                                                    |
|---------------------------|--------|------------------------------------------------------------------------------------------|
| AnimalSpecies.swift       | Modify | Expand tierNames to 15; replace maxAchievableOrderTier; add animalTierAppearance(tie     |
| ItemChain.swift           | Modify | Update makeAnimalChain loop; add .subObject/.powerUp; add makeSubObjectChain; reg        |
| GameStore.swift           | Modify | Bump to v21; add migrateV20toV21 (inject pityStates: [:], powerUpInventory: [])          |
| MergeBoardViewModel.swift | Modify | Update activateProducer for sub-object drops; power-up drag-apply handler; extend sellVa |

| File                               | Action     | Notes                                                                         |
|------------------------------------|------------|-------------------------------------------------------------------------------|
| AdoptionBoard.swift                | Modify     | Replace RescueStage-based weighting with direct tier-index weighting          |
| SanctuaryMap.swift                 | Modify     | Add 9 new SanctuaryArea entries for remaining families                        |
| InventoryStore.swift               | Modify     | Add powerUpInventory slot management                                          |
| SubObjectSystem.swift              | CREATE NEW | All pity/drop logic lives here                                                |
| ProducerTile (AnimalSpecies.swift) | Modify     | Add speedBurstActive, speedBurstRemaining, nextDropGuaranteedHighTier         |
| QuestGoal call sites               | Modify     | Reroute animalTierLabel/Symbol/Color to animalTierAppearance(tier:) — 3 sites |

## C. Phased Implementation Plan

### Phase 1 — 15-Tier Animal Chains (foundation, no new content type)

#### What gets built:

- AnimalSpecies.tierNames expanded to 15 entries per family (all 15 families × 15 stage names).
- ContentRegistry.makeAnimalChain loop changed from zip(RescueStage.allCases, tierNames) to tierNames.enumerated().
- animalTierAppearance(tier: Int) added; QuestGoal.animalTierLabel/Symbol/Color rerouted.
- maxAchievableOrderTier updated to cover tiers 0–14.
- AdoptionBoard.generateOrder weighting table updated for 15 tiers.
- sellValue(forTier:) extended to 15 entries.
- GameStore.currentVersion bumped to 21; migrateV20toV21 adds pityStates: [:].

**Files:** AnimalSpecies.swift, ItemChain.swift, AdoptionBoard.swift, MergeBoardViewModel.swift, GameStore.swift

**Player-facing result:** Animals merge through 15 stages with proper names. Adoption orders and quests reference correct tier labels. Existing tiers 0–8 continue to work — they just aren't at max tier anymore.

**Stable before next phase:** All persistence tests pass. Fresh game and loaded save display tier names at 0–14. Ambassador celebration fires at chain.maxTier (now tier 14).

### Phase 2 — Sub-Object Chain Data + Registry

#### What gets built:

- ChainCategory gains .subObject and .powerUp cases.
- SubObjectSystem.swift created with SubObjectRarity, PowerUpEffect, SpawnDropResult, SubObjectDropConfig, PityState.
- ContentRegistry.makeSubObjectChain() added; all 15 sub-object chains registered.
- GameState gains pityStates: [String: PityState] and powerUpInventory: [BoardItem?].
- InventoryStore gains powerUpInventory slot management and routing rule for .powerUp items.

**Files:** ItemChain.swift, SubObjectSystem.swift (new), GameStore.swift, InventoryStore.swift

**Player-facing result:** Sub-object chains exist in the registry. Power-up inventory exists in the UI. No actual drops yet.

**Stable before next phase:** Build green. ContentRegistry contains 15 animal + 15 sub-object chains. Round-trip save/load with pityStates and powerUpInventory passes persistence tests.

## Phase 3 — Sub-Object Drop Logic in Family Spawners

### What gets built:

- MergeBoardViewModel.activateProducer updated: familySpawner calls SubObjectSystem.resolveSpawnerDrop() to determine output.
- PityState per-species counters tracked in GameState.pityStates, updated after every spawn.
- Sub-object items placed on board exactly like animal items (same BoardItem model, different chainID).
- Sub-object merge logic works automatically (existing merge code is chain-agnostic).
- When a sub-object reaches tier 3 (power-up), it routes to powerUpInventory.
- Drop rate display added to spawner info text in selectedItemInfo.

**Files:** MergeBoardViewModel.swift, SubObjectSystem.swift, GameStore.swift

**Player-facing result:** Tapping a family spawner has a chance to produce sub-objects. Merging through 4 stages produces a power-up. Pity timers prevent long streaks without Rare/Epic drops.

**Stable before next phase:** All 15 families produce correct sub-object types. Drop rates observable in debug. Pity timers advance and reset correctly across sessions.

## Phase 4 — Power-Up Application Mechanics

### What gets built:

- ProducerTile gains speedBurstActive: Bool, speedBurstRemaining: Double, nextDropGuaranteedHighTier: Bool (backward-compatible via decodeIfPresent).
- tickProducers() extended to decrement speedBurstRemaining and clear speedBurstActive at zero.
- activateProducer respects speedBurstActive (halve cooldown) and nextDropGuaranteedHighTier (force higher tier, then clear flag).
- New gesture: dragging a power-up from powerUpInventory onto a spawner calls SubObjectSystem.applyPowerUp(effect:to:).
- CellView: visual indicator (glow or badge) when speedBurstActive or nextDropGuaranteedHighTier is active.

**Files:** AnimalSpecies.swift (ProducerTile), MergeBoardViewModel.swift, SubObjectSystem.swift, CellView.swift

**Player-facing result:** Players drag power-ups onto spawners for immediate effects. Speed Burst visibly accelerates the spawner ring. High-Tier Guarantee produces a better drop on the next spawn.

**Stable before next phase:** All four power-up effects apply correctly. speedBurstRemaining decrements in real time and across offline progress. Save/load preserves active buffs.

## Phase 5 — Map Area Expansion (9 remaining families)

### What gets built:

- 9 new SanctuaryArea entries in SanctuaryMap.swift: Ursids, Aquatics\*, Amphibians, Marsupials, Primates, Equines, Pachyderms, Bovines, Cervids.
- Each area has a newFamilySpawner reward and 2 upgrade tiers with UpgradeBonus passives.
- Build cost scaling: later areas cost higher-tier materials in larger quantities.
- \* Aquatics: path decision required (see Section E, Q4) — milestone vs. map area vs. both.

**Files:** SanctuaryMap.swift, MergeBoardViewModel.swift (if Aquatics milestone changes), AnimalSpecies.swift (if new UpgradeBonus fields needed)

**Player-facing result:** All 15 families unlockable through the map. Clear late-game build path beyond the current 6 areas.

**Stable before next phase:** Each new area's spawner, chain unlock, and upgrade bonuses work end-to-end. No family spawner appears before its area is built.

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## D. Risk & Conflict Assessment

### D.1 Energy wall / kibble economy

With 15-stage chains, players need ~75% more merges per ambassador. If kibble cost and regen rate are unchanged, progress will slow significantly. **Recommendation:** Either (a) increase the regen cap or reduce spawn cost for higher-multiplier modes, or (b) accept the slower pace as intentional. This is a balance decision — must be made before Phase 3 ships.

### D.2 Existing inventory logic

`InventoryStore.addItem()` currently routes `.tool/.material` to `toolInventory`, everything else to `inventory`. Adding `.subObject` and `.powerUp` requires two new routing cases. Sub-objects can default to the animal inventory; power-ups must have a dedicated destination.

### D.3 Current quest system

`QuestGoal.reachTier(.animal, tier: N)` where  $N \geq 9$  produces a fallback label (not a crash). Fixed in Phase 1 by `animalTierAppearance(tier:)`. No saved quests reference tiers  $\geq 9$  in the wild — no migration needed.

### D.4 Ambassador / star system

`MilestoneManager` hardcodes thresholds at 5/15/30/50. With 15-stage chains, reaching top tier is significantly harder — thresholds may need raising. The Aquatics 50-star side effect conflicts with the Phase 5 map area (see Q4 in Section E). The code handles a double-trigger safely but the intent needs clarification.

### D.5 Save data migration

Existing v20 saves with tiers 0–8 remain valid — they just represent a position in a longer chain. No tier-shifting migration needed. v21 migration is purely additive (inject `pityStates: []`, `powerUpInventory: []`). `ProducerTile` new fields use `decodeIfPresent` — backward-compatible.

### D.6 Board size and 15-stage chain depth

$9 \times 7 = 63$  cells. Reaching tier 14 theoretically requires  $2^{14}$  spawns, though players merge continually. Board management becomes the central skill. The existing 18-slot inventory may be insufficient for 15-stage chains across 15 families — see Section E, Q6.

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## E. Open Questions

The following require decisions before the affected phases can be implemented.

### Q1 (Phase 1 blocker) — Does the existing 9-tier chain get replaced or extended to 15?

The reference table's first two Canine stages match (Pup, Kit), then diverge. If existing stage names change, no tier-shifting migration is needed — only display strings change (from the registry, not saves). Players with saved tiers 0–8 will see those items renamed. If that's acceptable, no migration logic is needed at all.

## Q2 (Phase 1 blocker) — Kibble economy rebalancing for 15-stage chains.

Should the regen cap increase for late-game players? Should family spawners cost less kibble to incentivize exploring new families? Should the sub-object drop rate provide occasional free value to offset the longer grind?

## Q3 — Sell value scale for tiers 9–14.

Current scale tops at tier 8 → 500 coins. Geometric progression suggests 1,000 / 2,000 / 5,000 / 10,000 / 25,000 / 100,000 for tiers 9–14 — but this makes the top tier extremely valuable. Exact values need sign-off.

## Q4 (Phase 5 blocker) — Aquatics (.fish) — map area vs. milestone unlock.

Options: (a) keep milestone unlock, skip map area for Aquatics; (b) replace milestone unlock with map area; (c) keep both (milestone = early access, map area = upgrade). Decision needed before Phase 5.

## Q5 (Phase 3 blocker) — Pity timer thresholds.

What is the spawn count after which a Rare sub-object is guaranteed? After which an Epic? Suggested starting point: 30 spawns for Rare, 60 for Epic — but these need validation against your intended pacing.

## Q6 — Inventory expansion for 15-stage chains.

Do the current 18 animal inventory slots remain sufficient? The GDD v2.0 3-tab inventory is not yet coded. Should this be implemented alongside the chain expansion as a Phase 2 prerequisite?

## Q7 — Superpower system (GDD v2.1).

GDD v2.1 introduces per-family superpowers unlocking at Era 3 (Stage 7). Is this in scope for this implementation cycle, or deferred to a future Phase 6?

## Q8 (Phase 3 blocker) — Sub-object base drop rate from spawner.

Does the 60/25/10/5 split apply to every spawner tap (leaving 0% for animals), or does a configurable base rate (e.g. 20%) determine how often a sub-object drops at all, with the 60/25/10/5 split applying within that portion? The latter is recommended for balance.

## Q9 (Phase 5 blocker) — New upgrade bonuses for the 9 new map areas.

Existing UpgradeBonus struct has 12 fields, most already allocated. New areas may need new bonus types (sub-object drop rate, pity timer reduction, power-up duration). These require new fields in UpgradeBonus and handling in recalcActiveBonuses().

## Q10 — Board size confirmation.

GDD v2.0 specifies  $6 \times 5 = 30$  cells, but the code has  $9 \times 7 = 63$  cells. Confirming: the  $9 \times 7$  board is correct and the GDD needs updating, not the code.

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## Appendix: File Change Summary

| File                      | Phase(s) | Type   | Estimated Scope                                         |
|---------------------------|----------|--------|---------------------------------------------------------|
| AnimalSpecies.swift       | 1        | Modify | Large — data entry: 15×15 names + 15-tier helpers       |
| ItemChain.swift           | 1, 2     | Modify | Medium — loop change + new categories/builder           |
| AdoptionBoard.swift       | 1        | Modify | Small — tier weighting table                            |
| MergeBoardViewModel.swift | 1, 3, 4  | Modify | Medium — spawner drop path, power-up apply, sell values |
| GameStore.swift           | 1, 2     | Modify | Small — version bump + additive migration               |

| File                  | Phase(s) | Type       | Estimated Scope                                 |
|-----------------------|----------|------------|-------------------------------------------------|
| InventoryStore.swift  | 2, 4     | Modify     | Small — new category routing + powerUpInventory |
| SubObjectSystem.swift | 2, 3, 4  | CREATE NEW | Medium — drop logic, pity state, power-up apply |
| SanctuaryMap.swift    | 5        | Modify     | Large — 9 new area definitions + upgrade tiers  |
| CellView.swift        | 4        | Modify     | Small — speed burst / high-tier buff indicator  |